# Seemab — Mathematics Exam Overview

SINGLE ATTEMPT

exam-10-ch1-2-4-5-7-10 | Units 1, 2, 4, 5, 7, 10 (Full-Year Cumulative) | Federal Board Annual 2026
Date: 26 June 2026 | Total Marks: 65 | Time: 2 h 30 min

| 61/65 | 93.8% | A+ | 12/12 |
|---|---|---|---|
| TOTAL SCORE | PERCENTAGE | GRADE | MCQ (100%) |

## Section Breakdown

| | | |
|---|---|---|
| **Section A (MCQ)** | 12/12 | **100%** |
| **Section B (Short Q)** | 32/33 | **97.0%** |
| **Section C (Long Q)** | 17/20 | **85.0%** |

## MCQ Results (All 12 Correct)

| # | Topic | Correct | Seemab | Result |
|---|---|---|---|---|
| 1 | Irrational number ($\sqrt{7}$) | A | A | ✓ |
| 2 | Simplest form of $\sqrt{50}$ ($5\sqrt{2}$) | B | B | ✓ |
| 3 | $\log_{10} 1000 = 3$ | C | C | ✓ |
| 4 | $\log(mn) = \log m + \log n$ | B | B | ✓ |
| 5 | Factors of $x^2-7x+12$ | A | A | ✓ |
| 6 | $a^3+8$ sum of cubes | D | D | ✓ |
| 7 | Solve $2x+3=11 \rightarrow x=4$ | C | C | ✓ |
| 8 | $\lvert x\rvert=5 \rightarrow \{-5, 5\}$ | D | D | ✓ |
| 9 | Distance (0,0) to (3,4) = 5 | B | B | ✓ |
| 10 | Midpoint (2,4)&(6,10) = (4,7) | D | D | ✓ |
| 11 | Centre-to-circle = Radius | A | A | ✓ |
| 12 | Measurements for triangle = 3 | C | C | ✓ |

**7th consecutive perfect MCQ score (exams 14–20). Math MCQ all-time: 59/60.**

## Per-Question Summary

| Question | Variant | Topic | Marks | Earned | Notes |
|---|---|---|---|---|---|
| Q2(i) | OR | Rational/irrational classification | 3 | 3 | All three correct with reasons |
| Q2(ii) | OR | Scientific notation | 3 | 3 | Both correct ($4.6\times10^{-4}$ and $9.2\times10^{7}$) |
| Q2(iii) | OR | Expand $\log(a^2b/c)$ | 3 | 3 | All laws applied correctly |
| Q2(iv) | Main | log 6 and log 12 (given log 2, log 3) | 3 | 3 | Both correct |
| Q2(v) | Main | Factorize $x^2-9x+20$ and $2x^2+7x+3$ | 3 | 3 | Both factored correctly |
| Q2(vi) | Main | HCF of $12x^2y^3$ and $18x^3y^2$ | 3 | 3 | HCF = $6x^2y^2$ correct |
| Q2(vii) | Main | Solve $(2x)/3+1 = x/2+2$ | 3 | 3 | $x=6$ correct |
| Q2(viii) | Main | Solve $8x+9 < 6x-7$ | 3 | 3 | $x<-8$ correct with solution set |
| Q2(ix) | Main | Distance A(−2,1) to B(1,5) | 3 | 3 | 5 units correct |
| Q2(x) | Main | Midpoint (2,5)&(6,9) and quadrant | 3 | 2 | Midpoint correct; quadrant not stated (−1) |
| Q2(xi) | Main | Chord, tangent, arc definitions | 3 | 3 | All three definitions match key |
| Q3(i) | Main | Factorize $(x+1)\ldots(x+4)-24$; $\sqrt{4x^2+12xy+9y^2}$ | 5 | 5 | Perfect |
| Q3(ii) | Main | Log laws; evaluate log 42, log(9/2) | 5 | 5 | Perfect — log 42 = 1.6232, log(9/2) = 0.6532 |
| Q3(iii) | Main | Solve & verify $(x-2)/3+(x+1)/2=4$; inequality on number line | 5 | 4 | Stopped before finishing; $x=5$ and verification absent (−1) |
| Q3(iv) | Main | Square proof (sides + diagonals); distance formula | 5 | 3 | 4 sides correct; diagonal check absent (−2) |

**Performance Strengths**

- Perfect MCQ for the 7th consecutive exam
- Logarithm evaluation (Q3(ii)) and expansion (Q2(iii)) both perfect
- Substitution factorization of degree-4 product (Q3(i)) fully correct
- Log 42 and log(9/2) both evaluated correctly using the given values — no errors in either calculation
- All linear equations and inequalities in Section B answered correctly
- Coordinate geometry formulas applied cleanly with no arithmetic errors
- Practical geometry definitions match textbook language precisely
- HCF by prime factorization method shown in full

**Areas to Improve**

- **Square proof incomplete:** equal sides prove rhombus only; must also compute diagonals (AC = BD = √58) to confirm square — 2 marks lost
- **Solution left unfinished:** "solve and verify" requires both a final stated answer and explicit substitution back — stopped at 5x−1 = 24 without completing — 1 mark lost
- **Quadrant not stated:** after correct midpoint, forgot to write "Quadrant I" — 1 mark lost
- **Pattern (5th exam):** stopping before all required sub-parts / explicit instructions is the critical recurring weakness across all subjects

## Math Subject Performance History

| # | Date | Exam | Score | % | MCQ | Sec B | Sec C |
|---|---|---|---|---|---|---|---|
| 1 | 18 Apr 2026 | exam-01-ch1-2 | 65/65 | 100% | 12/12 | 33/33 | 20/20 |
| 2 | 22 Apr 2026 | exam-03-ch4-5-7 | 60.5/65 | 93.1% | 12/12 | 28.5/33 | 20/20 |
| 3 | 25 May 2026 | exam-04-ch4 | 57/65 | 87.7% | 12/12 | 26/33 | 19/20 |
| 4 | 5 Jun 2026 | exam-07-ch8 | 64/65 | 98.5% | 11/12 | 33/33 | 20/20 |
| **5** | **26 Jun 2026** | **exam-10-ch1-2-4-5-7-10** | **61/65** | **93.8%** | **12/12** | **32/33** | **17/20** |

Subject average after this exam: 307.5/325 = **94.6%**. Section C 17/20 (85%) is the lowest of the five Math exams, driven by the unfinished equation and absent diagonal check — both procedural rather than conceptual. MCQ returned to 12/12 after the single error on exam-07.

**Note for parent**

This is a strong overall performance (93.8%, A+) on a demanding full-year cumulative paper covering six units. The 4 marks lost are entirely from three procedural oversights, not from conceptual misunderstanding. Seemab clearly knows the mathematics: the losses are from not finishing a final algebraic step, not stating the quadrant, and not computing the diagonals in the square proof. These are all fixable with a simple habit of re-reading each question before moving on. The logarithm section was handled with complete confidence — both log 42 and log(9/2) fully and accurately derived.